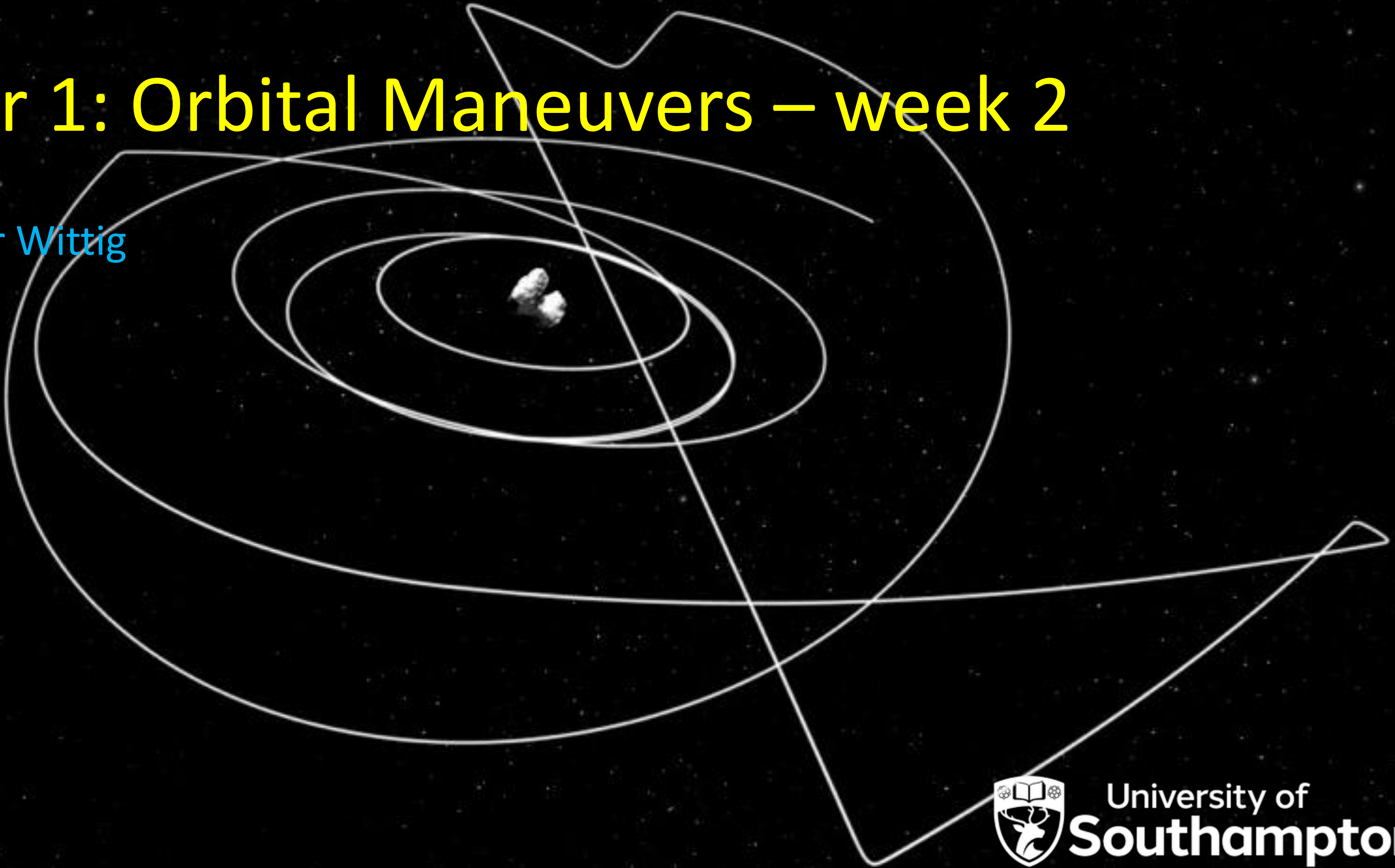
Orbital Mechanics (SESA6076)



Rosetta lander deployment orbit (ESA)

Chapter 1: Orbital Maneuvers – week 2

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Questions

- Any questions on previous weeks content?

Course Content

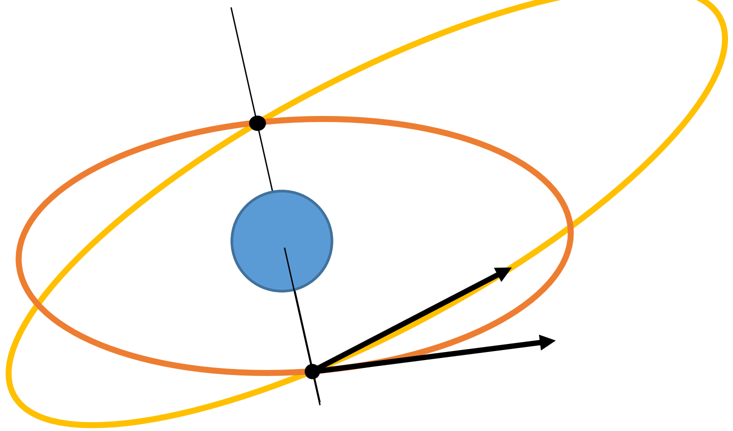
- **Introduction**
- **Chapter 1: In 20.000 km, turn right.**
  Orbital Manoeuvres
  - Coplanar Manoeuvres
  - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing**
  Orbital Perturbations & Propagation
  - Non-Uniform Gravity Field
  - Third-Body Perturbation
  - Atmospheric Drag
  - Solar Radiation Pressure
  - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird?**
  Orbit Observation & Determination
  - Observation Techniques
  - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda**
  Relative Orbital Motion
  - Relative Orbital Motion
  - Linearization & State Transition Matrix
- **Chapter 5: To boldly go…**
  Interplanetary Trajectories & Optimization
  - Lambert's Problem
  - Patched Conics
  - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies**
  Restricted Three-Body Problem
  - N-Body Problem
  - Circular Restricted Three-Body Problem
  - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round**
  Attitude Dynamics
  - Attitude representation
  - Attitude Dynamics
- **Chapter 8: In the spin cycle**
  Attitude Motion
  - Torque-Free Motion
  - Attitude Maneuvers
- **Summary and Revision**

Plane Change Maneuvers

Plane change

- So far all maneuvers were planar (2D)

- Same principle in 3D
  - Find maneuver point (fixes rotation axis)
  - Find old and new velocity vectors
  - Calculate vectorial difference and ΔV



Plane change

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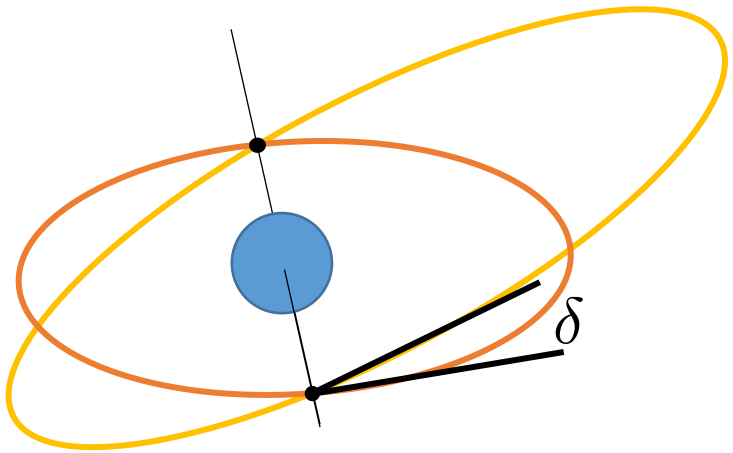

- Maneuvers changing orbital plane sometimes also called “inclination change” maneuver
  - **NOT** the same as Keplerian inclination *i* or flight path angle
  - Need angle δ between orbital planes e.g. from angular momentum vectors.
  - Example: Walker constellations

$$\cos(\delta) = \frac{\vec{h}_1 \cdot \vec{h}_2}{h_1 h_2}$$

Recall: How to arrange satellites in constellation?

- Walker delta pattern:

  i: t/p/f

  - i: inclination of orbital planes
  - t: total number of satellites
  - p: total number of planes
  - f: phase shift of satellites in adjacent planes

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Recall: Walker constellations

Recall: How to arrange satellites in constellation?

- Walker delta pattern:

  i: t/p/f

  - i: inclination of orbital planes
  - t: total number of satellites
  - p: total number of planes
  - f: phase shift of satellites in adjacent planes
- All orbits are similar
  - Simpler maintenance
  - No relative drift
  - Same mission constraints

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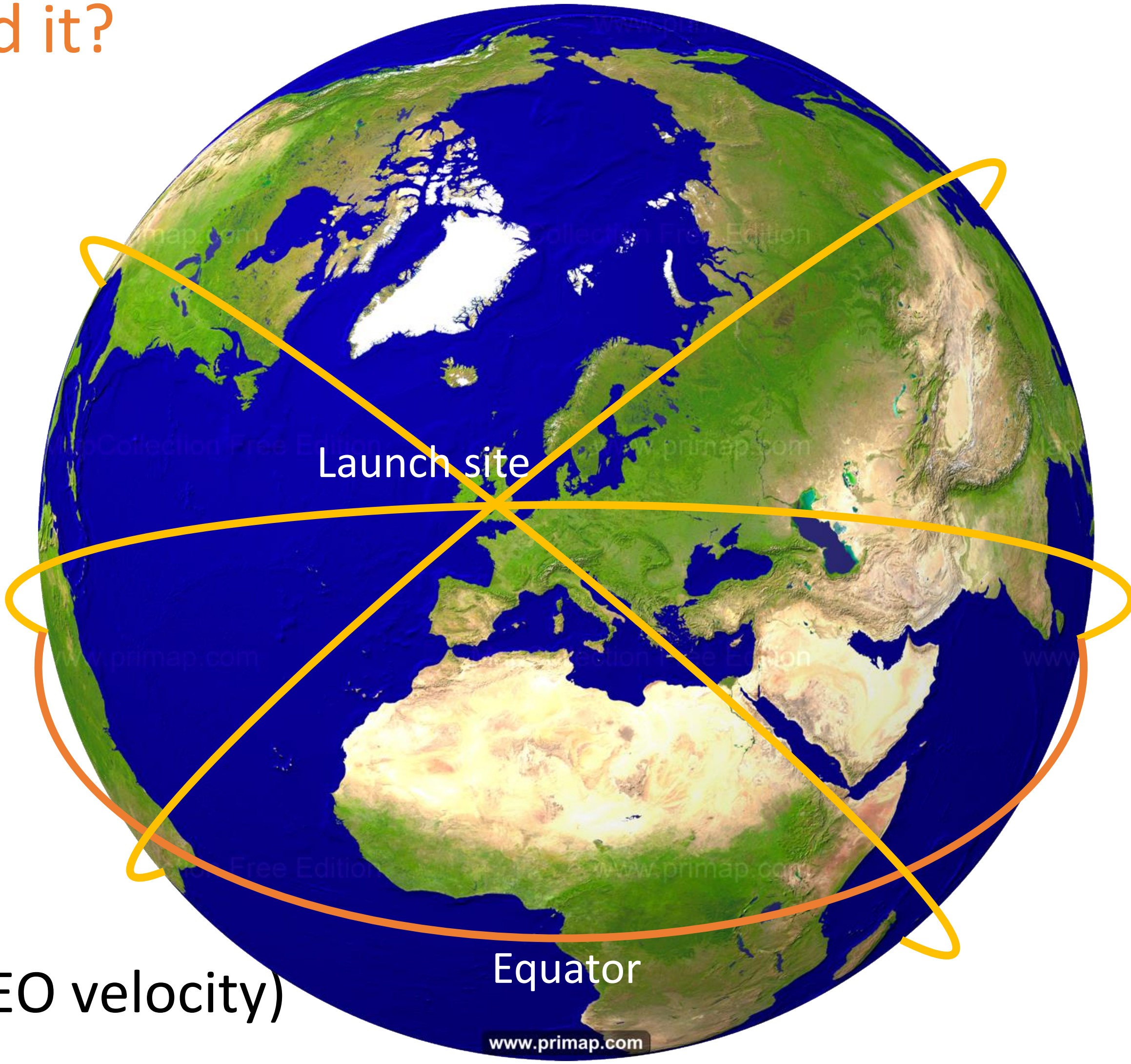
Plane change is expensive! Why do we need it?

- For Earth satellites:
  - Coverage
  - Observation constraints
  - Orbit evolution

Why not launch directly into correct plane?

- Launch constraints!
  - Can only launch into inclination larger than launch site latitude!
  - Use Earth rotation to reduce launch cost (0.46 km/s at equator, 5%-10% LEO velocity)



Gwynne Shotwell

Career:

- MSc in Applied Mathematics (Northwestern University)
- Researcher (The Aerospace Corp)
- Director of Space Systems (Microcosm Inc)
- President & Chief Operating Officer (SpaceX, 2002)

Achievements:

- Runs SpaceX day-to-day business
  $5bn in contracts, 3000+ people on payroll
- 59th most powerful woman in the world (Forbes, 2018)


SpaceX

Rotation of orbital plane

Rigid rotation of orbit, **no shape change**

- Rotation axis is radial vector
- Split velocity vector into radial and horizontal:

$$\vec{v} = \vec{v}_r + \vec{v}_\perp$$

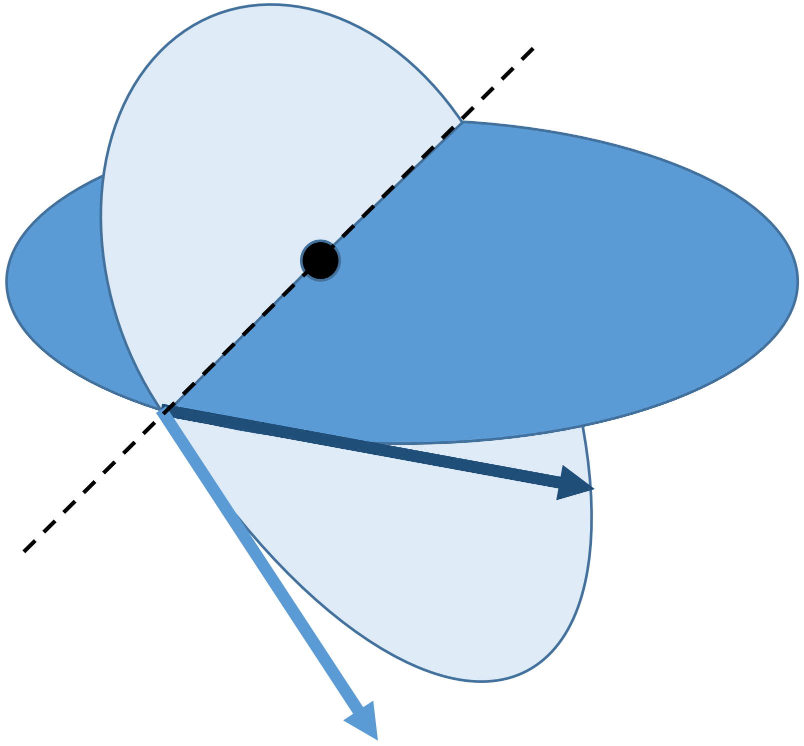
- Rotation only acts on $\vec{v}_\perp$:

$$\vec{v}^*_\perp = Rot_r(\delta)\vec{v}_\perp$$

- Resulting velocity: $\vec{v}^* = \vec{v}_r + \vec{v}^*_\perp$

- Required Δv: $\boxed{\Delta v = \left|\vec{v}^* - \vec{v}\right| = \left|\vec{v}^*_\perp - \vec{v}_\perp\right|}$

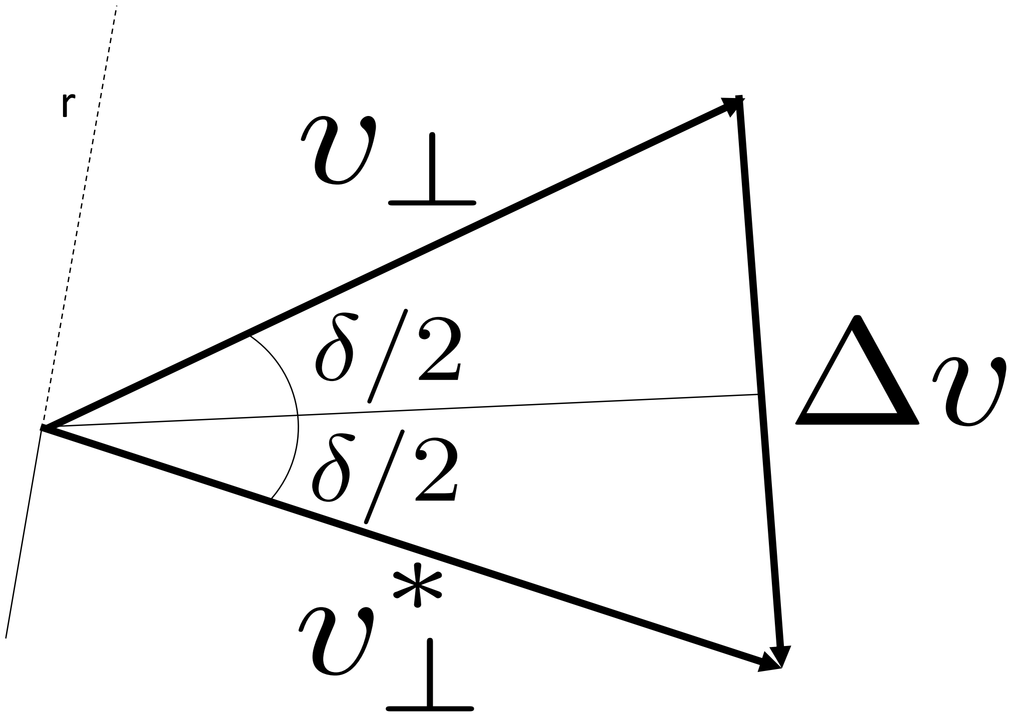
2D problem in horizontal plane!



Recall: v_r and $v_\perp$ from orbit shape and true anomaly.

Rigid rotation of orbit, **no shape change**

- In horizontal plane:



- From triangle: $\Delta v = 2v_\perp \sin(\delta/2)$
- Note: $v_\perp = v_\perp^*$ because **rigid** rotation

Cost of rotation proportional to horizontal velocity!

Example

After a satellite is launched into a 400km altitude circular orbit from Cape Canaveral (lat N28.46^o), what is the minimum Δv required to reduce the inclination of the orbit to zero?

$\mu = 398600\ km^3/s^2$

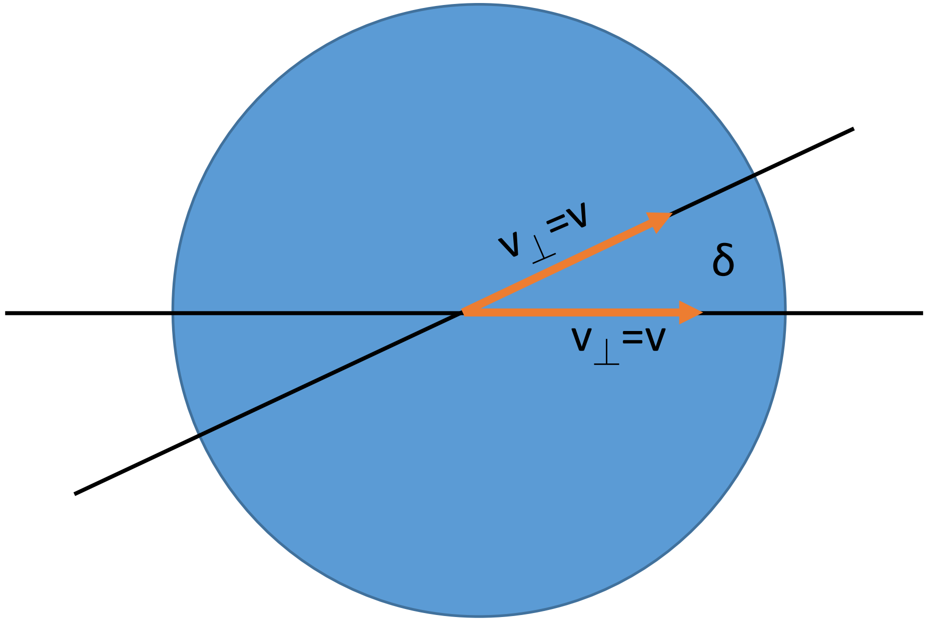
$R_E = 6371\ km$

Example

After a satellite is launched into a 400km altitude circular orbit from Cape Canaveral (lat N28.46^o), what is the minimum Δv required to reduce the inclination of the orbit to zero?

δ=28.46 (minimum possible for this launch)
μ=398600 km^3/s^2
r = 6371 km + 400 km

Δv = 3.77 km/s



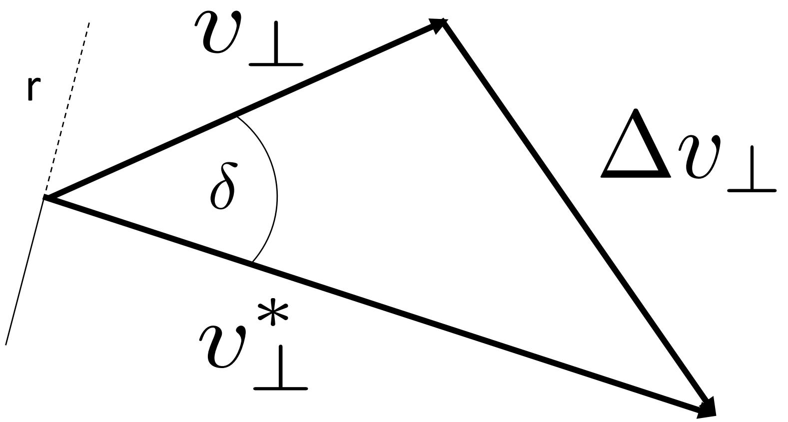
(side view)

General Δv calculation

- Same derivation allows to generalize for any change in velocity:

$$\Delta v = \sqrt{(v_r^* - v_r)^2 + {v_\perp^*}^2 + {v_\perp}^2 - 2v_\perp^* v_\perp \cos(\delta)}$$

  - Different radial velocity
  - Different horizontal velocities (magnitude & direction)



- In terms of flight path angles:

$$\Delta v = \sqrt{{v^*}^2 + v^2 - 2v^* v[\cos(\gamma^* - \gamma) - \cos(\gamma^*)\cos(\gamma)(1 - \cos(\delta))]}$$

Special cases

$$\Delta v = \sqrt{(v_r^* - v_r)^2 + {v_\perp^*}^2 + {v_\perp}^2 - 2v_\perp^* v_\perp \cos(\delta)}$$

$$\Delta v = \sqrt{{v^*}^2 + v^2 - 2v^* v[\cos(\gamma^* - \gamma) - \cos(\gamma^*)\cos(\gamma)(1-\cos(\delta))]}$$

- Orbit plane rotation around common apse line:

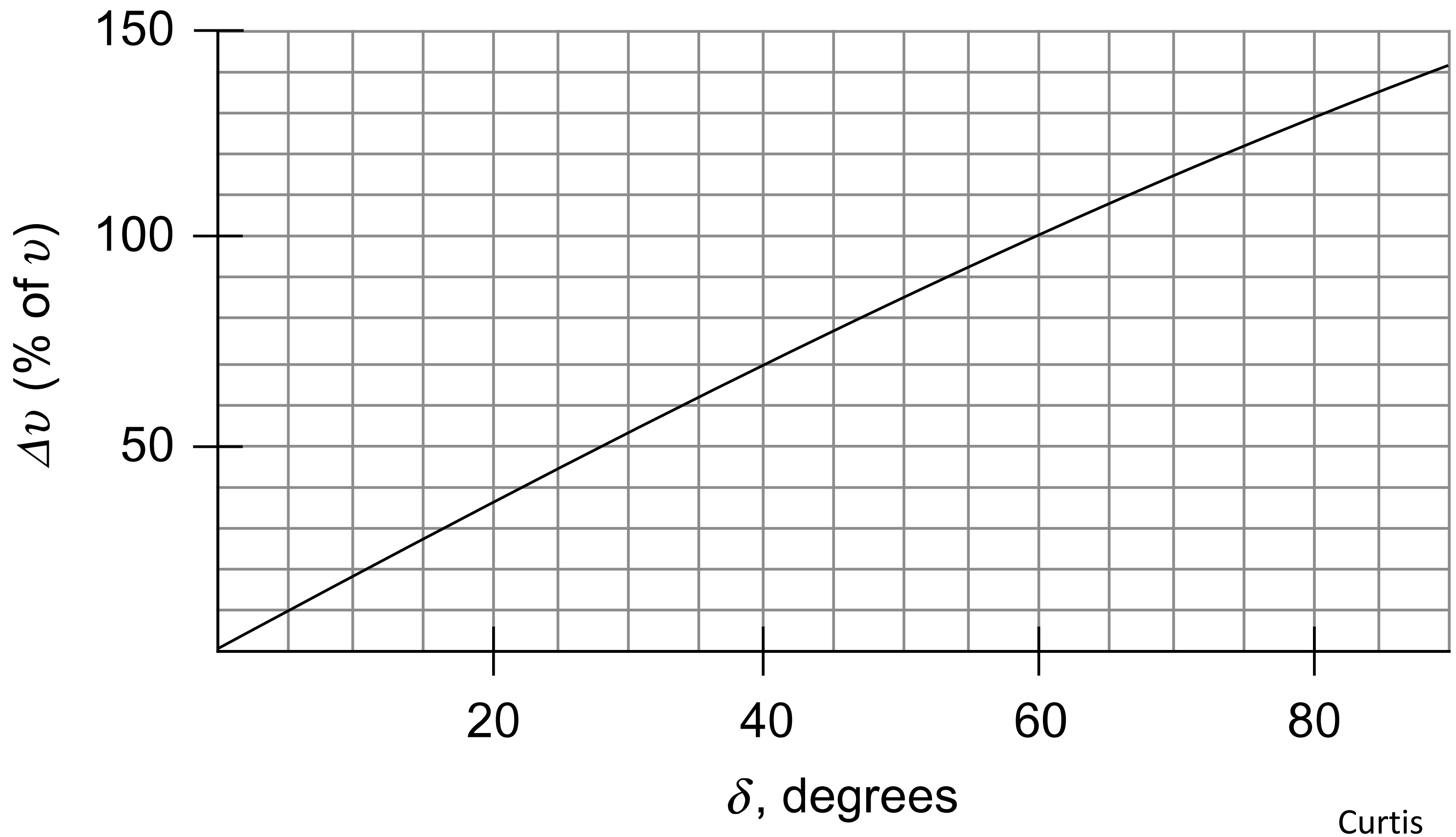
$$v_r = v_r^* = 0 \Rightarrow \Delta v = \sqrt{{v^*}^2 + v^2 - 2v^* v\cos(\delta)}$$

- No plane change:

$$\delta = 0 \Rightarrow \Delta v = \sqrt{{v^*}^2 + v^2 - 2v^* v\cos(\gamma^* - \gamma)}$$

Same equation we derived last week for co-planar case

Plane change is expensive



from equatorial to near-polar orbit: ~140%

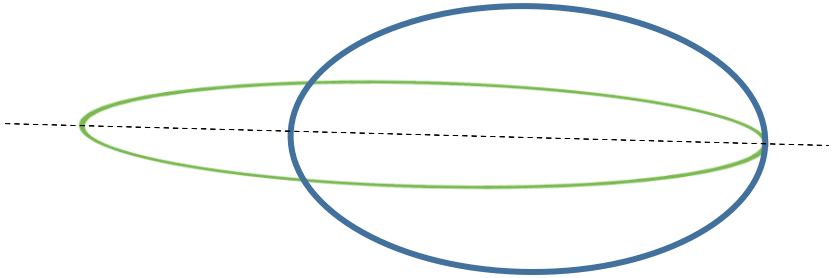
Curtis

Cost of a plane change of given angle as percentage of total horizontal velocity

Plane change is expensive

Always combine maneuvers at the same point!

Triangle inequality!



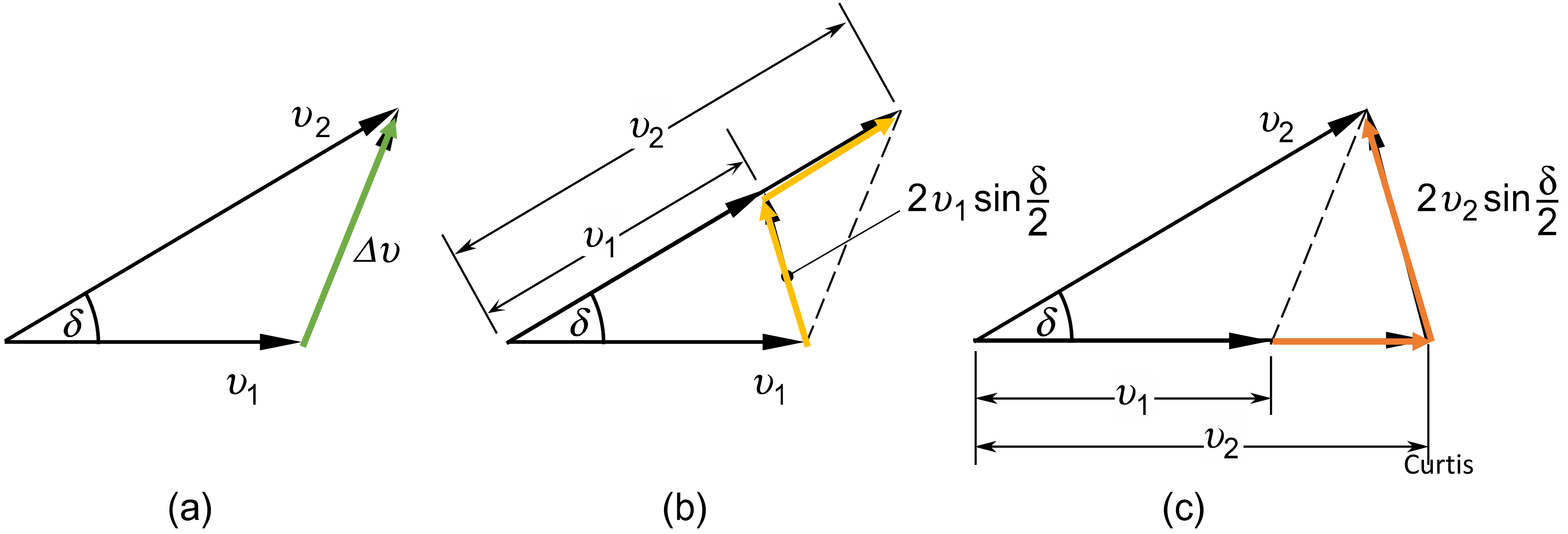


FIGURE 6.24

Orbital plane rotates about the common apse line. (a) Speed change accompanied by plane change. (b) Plane change followed by speed change. (c) Speed change followed by plane change.

Optimal maneuver location

How to reduce the cost of orbital plane changes?

Plane change cost proportional to $v_\perp = h/r$.
So: reduce $v_\perp$!

$$\Delta v = 2 v_\perp \sin(\delta/2)$$

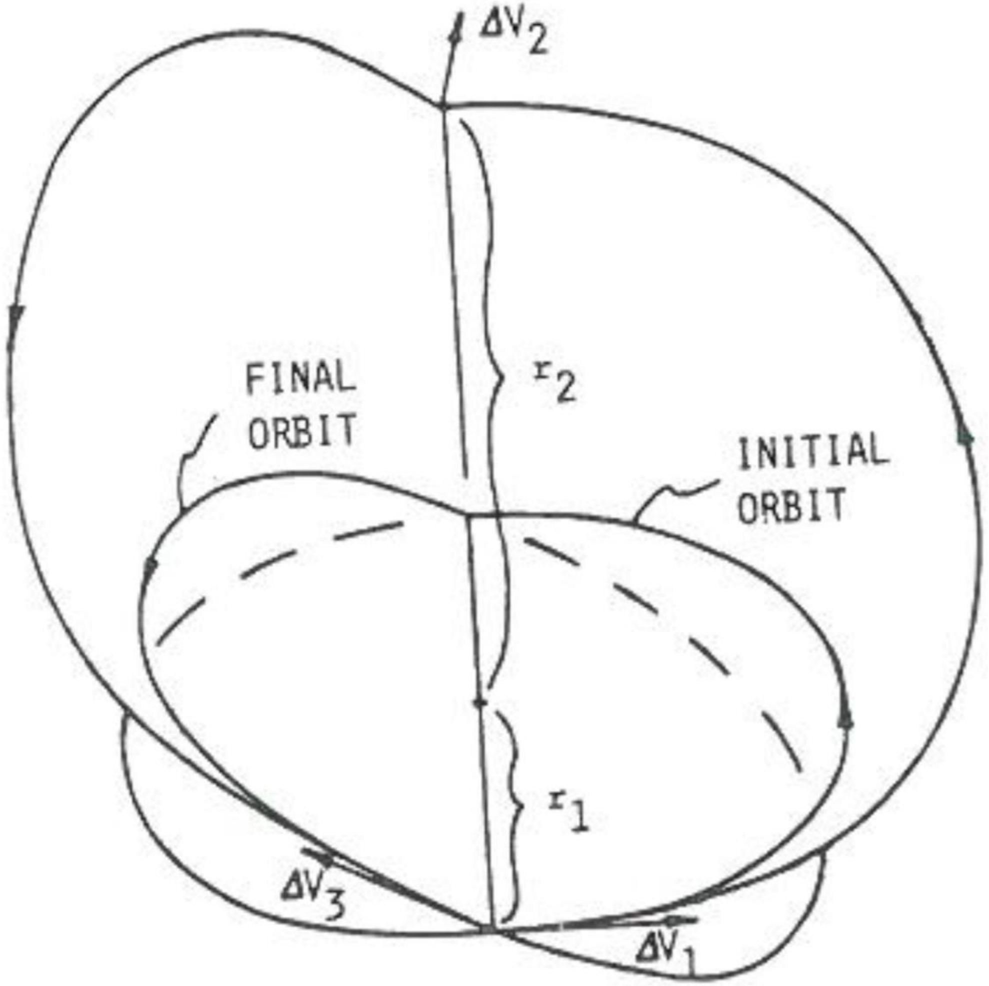
Inclination change **cheapest** when done far from primary (i.e. apoapsis).

- As with planar transfers: clever multi-burn maneuvers can help
- Example: Restricted three impulse plane change
  - First planar burn to raise apoapsis
  - Second (cheap) inclination change burn
  - Third (new) planar burn to lower apoapsis again

Restricted 3 impulse circular plane change

Goal: Change inclination of **circular** orbit by δ

1. Tangential burn to transfer ellipse with apoapsis r_2
2. At apoapsis: inclination change by rigid rotation
3. At periapsis: tangential burn back to circular orbit



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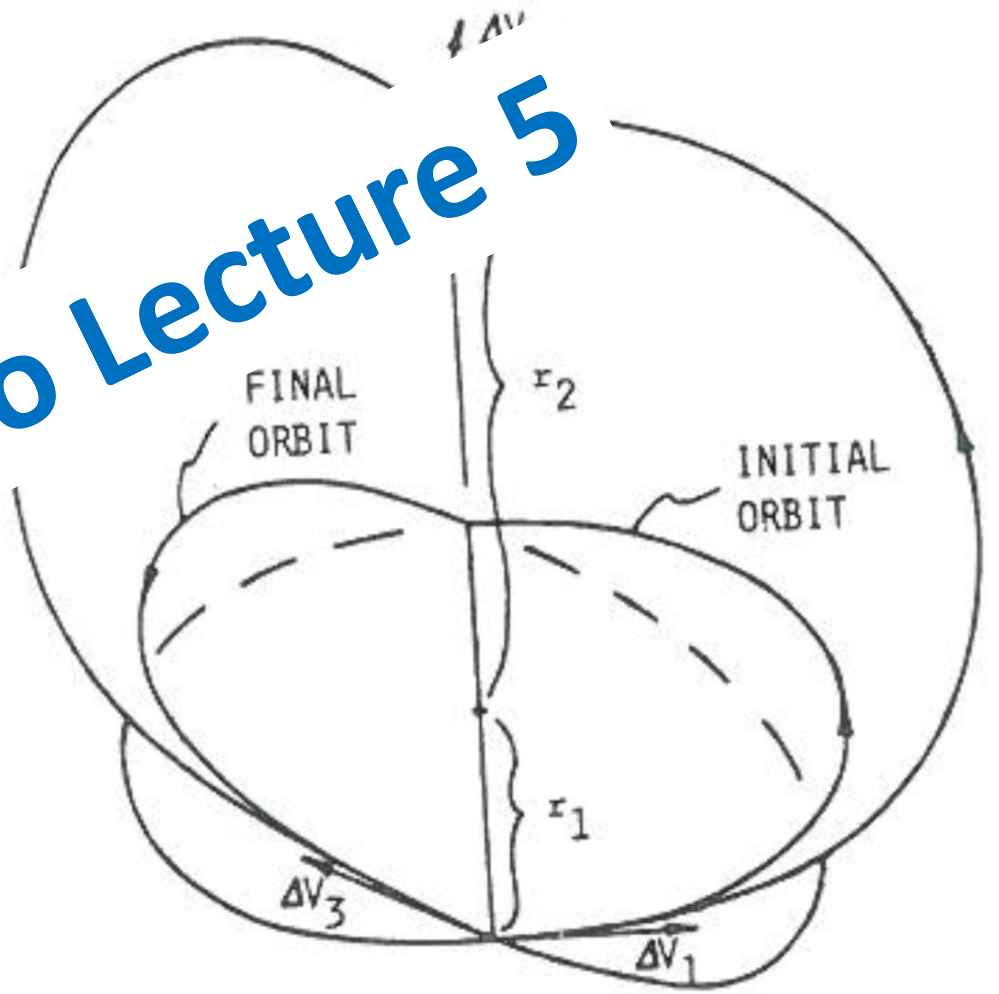
Restricted 3 impulse circular plane change

Goal: Change inclination of **circular** orbit by δ

1. Tangential burn to transfer ellipse with apoapsis r_2
2. At apoapsis: inclination change by rigid rotation
3. At periapsis: tangenti-' back to circular -

see Chapter 1 Video Lecture 5



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Oberth Effect

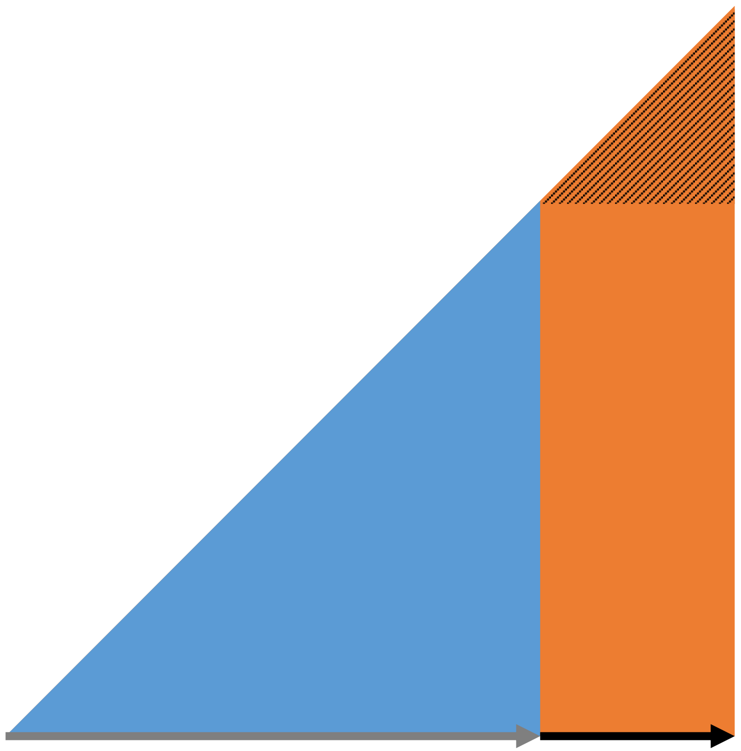
Oberth Effect: Impulsive maneuver at high speed maximizes energy gain.

Example: parabolic escape trajectory + Δv

	Δv @ Infinity	**Δv @ Perigee**
Perigee	$\epsilon = \frac{1}{2} v_{esc}^2$	$\epsilon = \frac{1}{2} \left(v_{esc} + \Delta v\right)^2 = \frac{1}{2} v_{esc}^2 + v_{esc}\Delta v + \frac{1}{2}\Delta v^2$
Infinity	$\epsilon = \frac{1}{2}\Delta v^2$	$\epsilon = \frac{1}{2}\Delta v^2 + v_{esc}\Delta v$

Hermann Julius Oberth (Wikipedia)

Optimal maneuver location



Where does extra energy come from?

- Potential energy of propellant
- Kinetic energy in exhaust

Where should maneuvers take place?

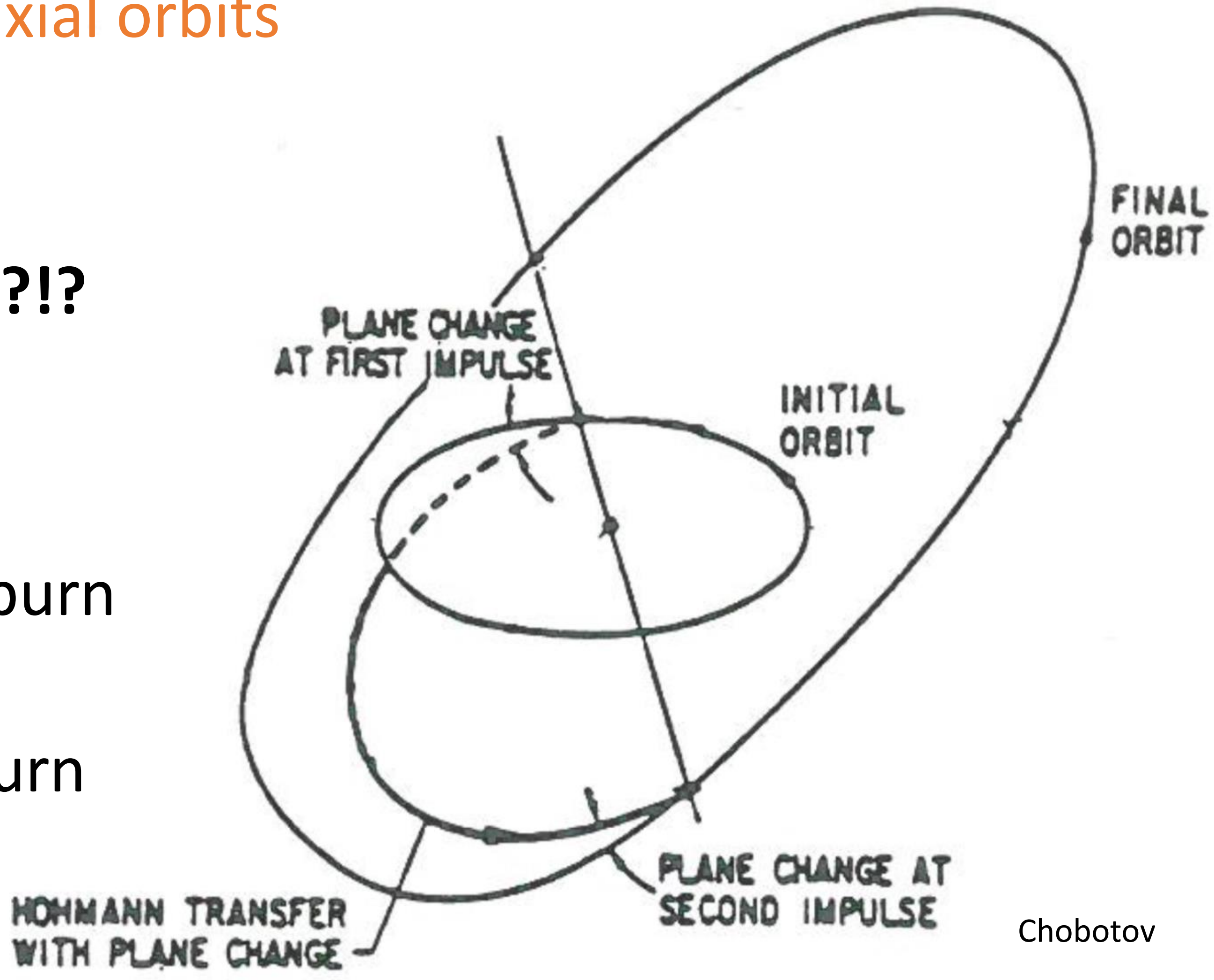
- Inclination change: as far as possible
  - Rotating smaller velocity
  - Free at infinity
- Energy gain: as close as possible
  - Higher velocity
  - Pump up semi-major axis

Can be beneficial to move before maneuvering!

Goal: Hohmann transfer between co-axial orbits with rotation around apse line.

What if Oberth and plane change mix?!?

- Hohmann has two burns
- Can perform plane change at either burn
- Most general: split plane change
  α : fraction of plane change at first burn



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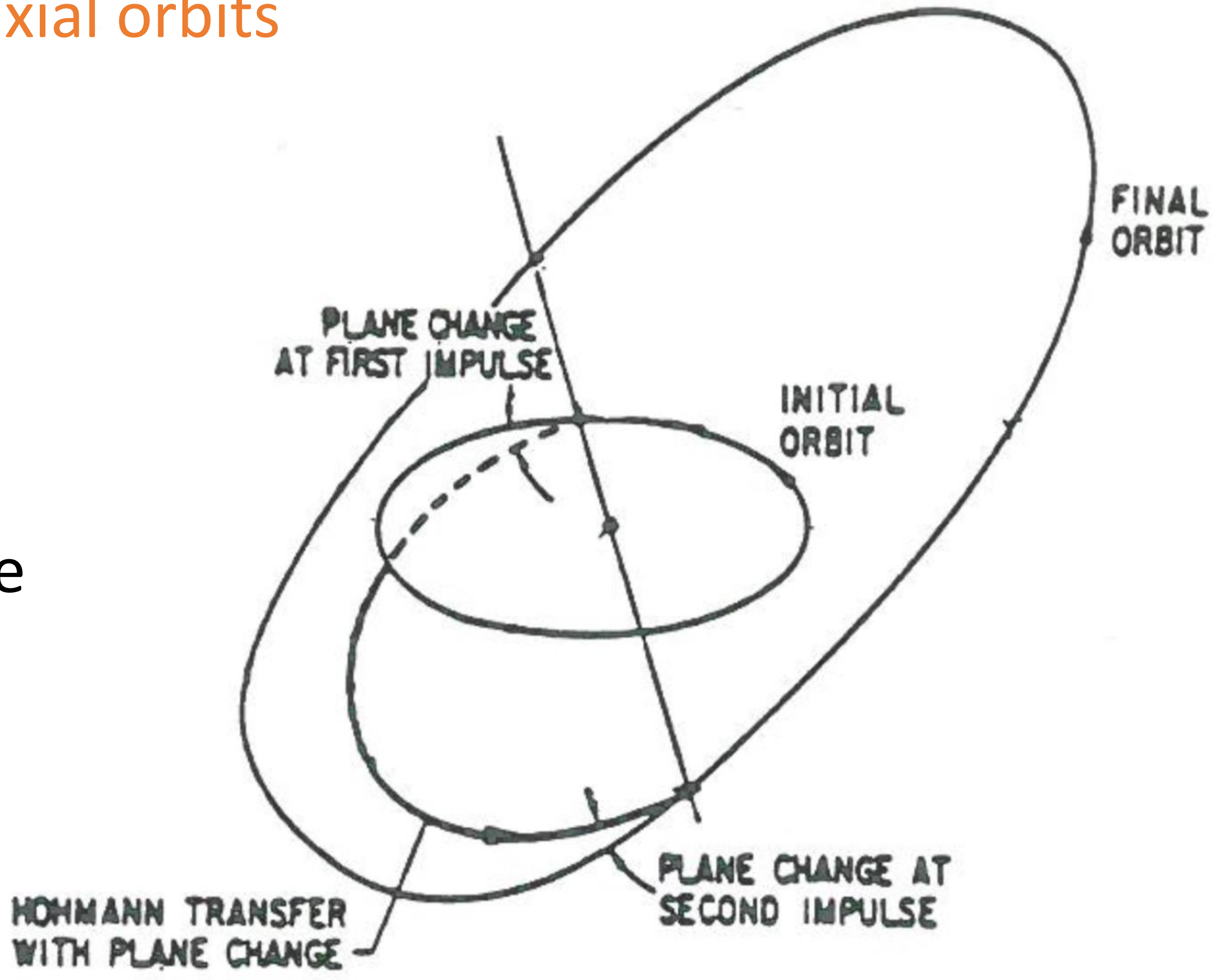
Hohmann transfer with split plane change

Goal: Hohmann transfer between co-axial orbits with rotation around apse line.

- Can minimize Δv w.r.t. α
  - Set derivative to zero
- Resulting equation can for α_{opt} can be solved numerically

Past exam question:
Derive expressions for Δv and the optimality condition (without solving it)



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Goal: Hohmann transfer between co-axial orbits with rotation around apse line.

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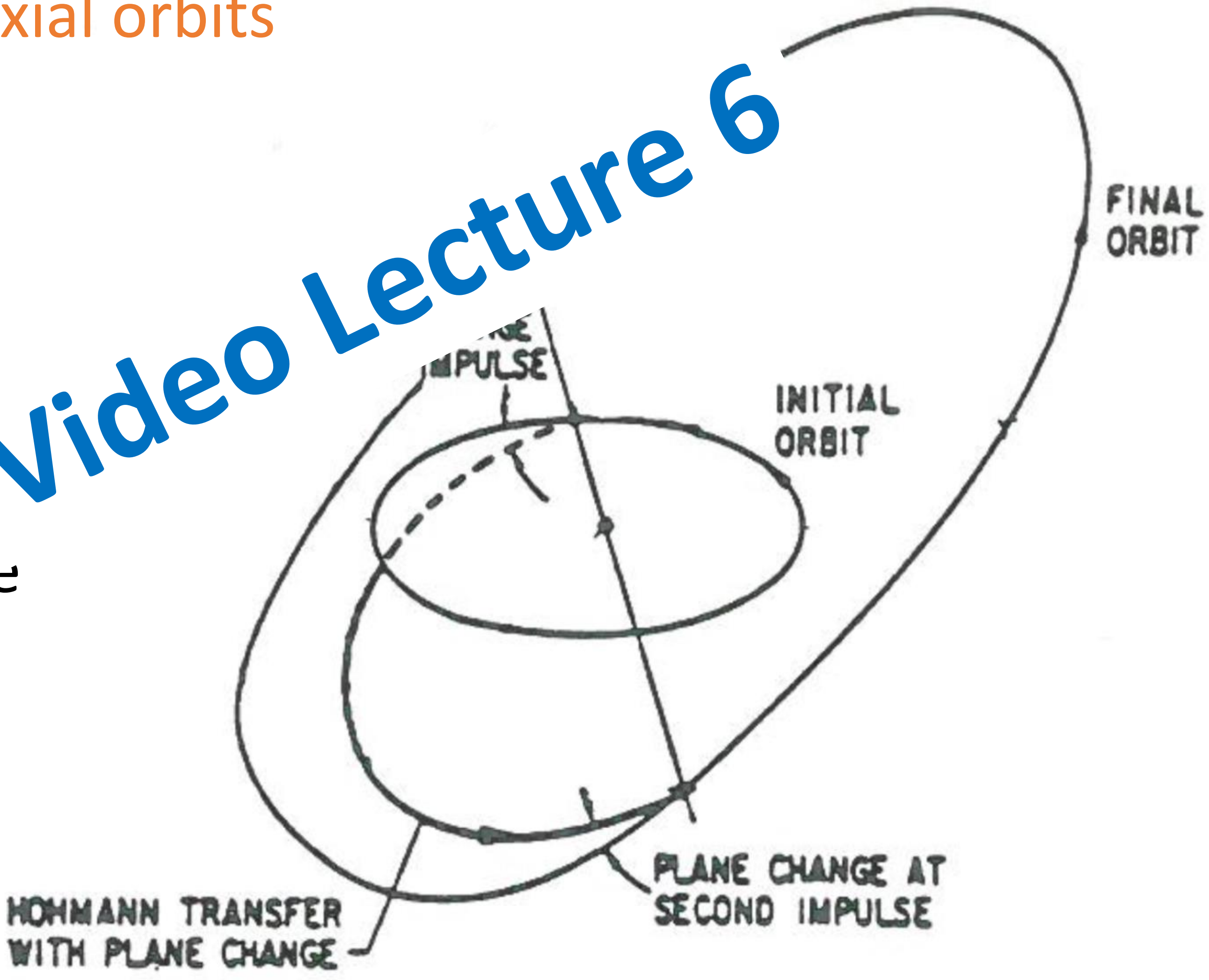
  - Set derivative to zero

- Resulting equation car ... e solved numericall...

see Chapter 1 Video Lecture 6

Past exam qu...
Derive express... ...or Δv and the optimality condition (without solving it)



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Summary

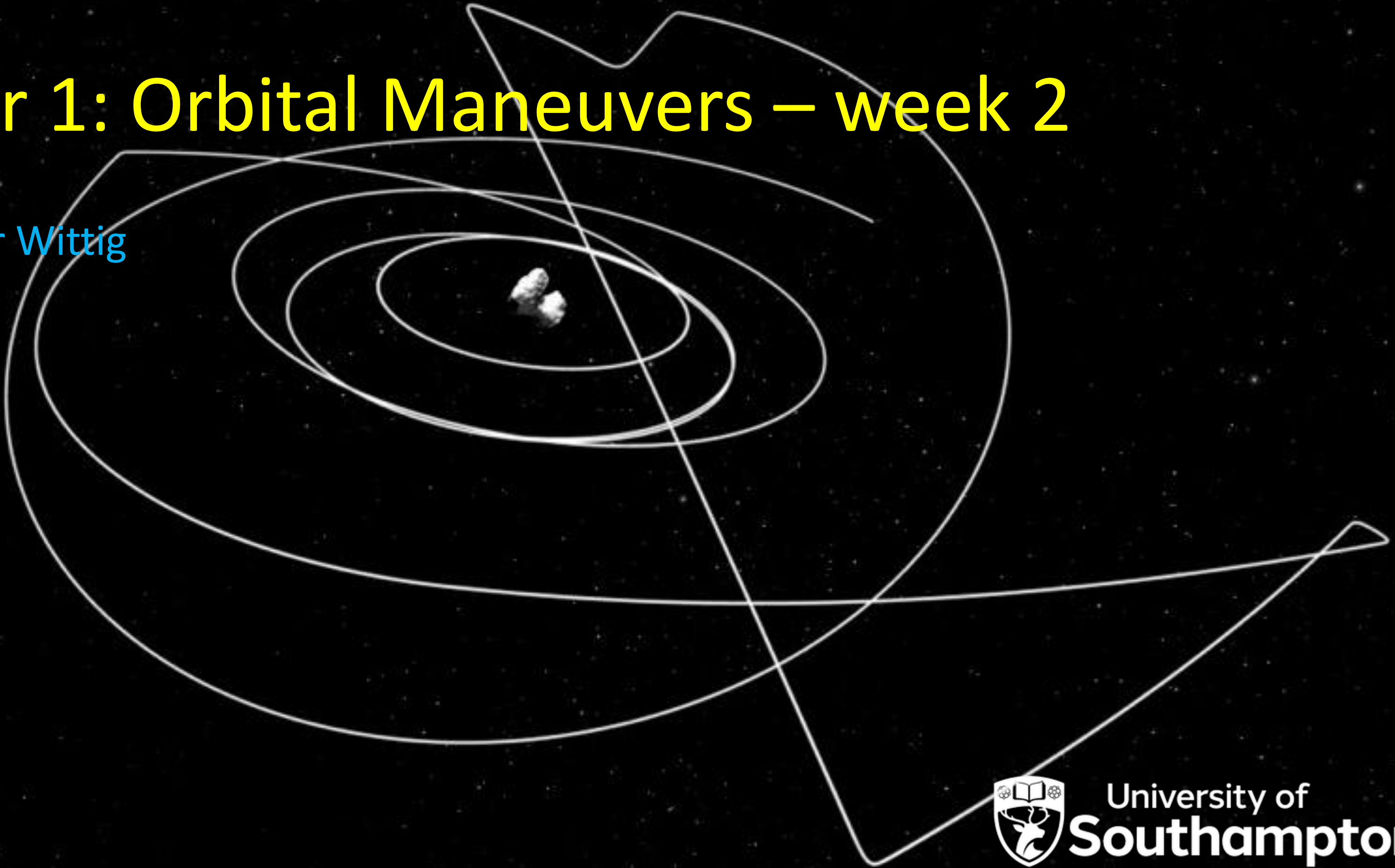
- Impulsive orbit transfers instantaneously change velocity
- Cost is given by the **vectorial** difference between initial and final velocity
- Multi-burn maneuvers can reduce the cost of maneuvers
- Plane change can be added in various ways to any planar maneuver
- Optimal maneuver often determined numerically
- Combined burns at the same point are always better than repeated burns

Further reading

- Curtis, Chapter 6
- Vallado, Chapter 6

Chapter 1: Orbital Maneuvers – week 2

Dr. Alexander Wittig



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